

28 — MKUFT Discriminating Experiments and Promotion Gates

Professional reading edition of the evolving public MKUFT canon

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1. Purpose

The existing experimental programme correctly identifies many places where MKUFT-related claims could be tested. This module adds a stricter distinction between:

- a known neighbouring effect;
- an exploratory anomaly assay;
- a theory-discriminating signature;
- a mechanism-level flagship test;
- a foundational recovery programme.

The governing correction is:

A positive anomaly is not an MKUFT result unless MKUFT predicted the pattern that distinguishes it from accepted alternatives before the result was known.

RNG bias, unusual noise, geometry dependence, observer effects, human–AI improvement, quantum contextuality, decoherence and information-dependent behaviour all have existing literatures. Reproducing one of those broad effect classes does not establish MKUFT. The framework earns support only when its typed variables, interactions, scaling rules, ablations or recovery conditions add reproducible predictive value.

This document therefore does not delete the broad assays in [04_EXPERIMENTAL_TEST_PROGRAM.md](#). It governs how they may be promoted.

2. Evidence tiers

Tier 0 — calibration or neighbouring established effect

A Tier 0 result shows that the apparatus, measure or task can detect a known effect.

Examples:

- geometry changes an ordinary resonance exactly as Maxwell, acoustic or mechanical models predict;
- attention changes a recognised physiological measure;
- reciprocal dialogue differs from AI-alone performance;
- a quantum measurement protocol reproduces established contextuality or Wigner-type constraints.

Tier 0 validates competence and instrumentation. It does not support MKUFT by itself.

Tier 1 — reconnaissance assay

A Tier 1 assay asks whether a candidate residual or interaction is present under strong controls.

It may estimate:

- variance;
- plausible effect size;
- measurement reliability;
- source stability;
- operational definitions;
- candidate interaction shape.

A Tier 1 positive result remains exploratory unless its primary statistic and predicted direction were locked before collection. A Tier 1 null can still remove an unproductive branch.

Tier 2 — discriminating signature test

A Tier 2 test predicts a structured fingerprint that competing models do not predict equally well.

A valid signature normally includes at least three of:

- source-class specificity;
- condition dependence;
- sign or ordering;
- scaling with a measured variable;
- threshold or reversal;
- absence in a matched negative control;
- deformation after a load-bearing component is removed;
- successful held-out prediction.

The primary object is an interaction or response surface, not an isolated p-value.

Tier 3 — mechanism flagship

A Tier 3 test supplies a parameterised mechanism or architecture that:

- beats the strongest fair ordinary null;
- predicts held-out results;

- survives independent replication;
- deforms in the predicted way when load-bearing relations are removed;
- returns to the accepted baseline when the additional MKUFT term is absent.

Tier 3 establishes only the mechanism-level result actually tested. It does not automatically prove the whole ontology.

Tier 4 — foundational recovery and discrimination

A Tier 4 programme must recover the established regime it claims to unify and then produce an operational difference.

For the physics branch this includes, where applicable:

- Bell-compatible correlations;
- no-signalling;
- Born-rule and quantum recovery;
- relevant quantum-field-theory limits;
- general-relativistic or experimentally equivalent gravitational recovery;
- dimensional consistency and conservation requirements;
- at least one new quantitative prediction or superior derivation.

A framework that only redescribes accepted results remains an interpretation or systems translation, not a new physical mechanism.

3. Promotion gates

An experiment should not be described as a flagship until it passes every applicable gate.

Gate 1 — exact object

State one claim in one declared domain. Do not test “MKUFT” as an undifferentiated whole.

Gate 2 — typed variables

Every independent variable, outcome, latent measure and coupling must have:

- a layer address;
- units or scale;
- a measurement procedure;
- a declared uncertainty model.

“Coherence,” “observer,” “information,” “geometry” and “boundary” are not measurements until operationalised.

Gate 3 — strongest fair null

Compare against the best ordinary explanation, not a weakened caricature.

The null set should include, where relevant:

- accepted physical dynamics;
- full device and environmental models;
- expectation and demand effects;
- classical stochastic sources;
- deterministic pseudorandom controls;
- independent ensembles;
- one-way assistance;
- strong alternative pairings;
- complete full-history replay or transfer conditions;
- flexible nonparametric residual models.

Where retained relational history is claimed to be load-bearing, the replay condition must give fresh, reset, or matched substitute systems the complete frozen explicit record, fair preparation time, tools, and opportunity for reciprocal work. Otherwise a deficit may show only missing information or familiarity rather than traversal-path dependence.

Gate 4 — locked prediction

Before confirmatory data are collected, specify at least:

- primary statistic;
- sign or ordering;
- expected functional form or bounded family;
- smallest effect of interest;
- exclusion and stopping rules;
- conditions where the effect should be absent.

If Stage 1 is used to identify these quantities, Stage 1 cannot also confirm them. Freeze the model and test it on independent Stage 2 data.

Gate 5 — interaction fingerprint

Prefer a predicted interaction over a broad main effect.

A finding such as “condition A differs from condition B” is weak when many mechanisms can produce it. A stronger result has the form:

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source class × information route × measured state
```

or:

```
parameter sweep × predicted residual shape × held-out platform
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Gate 6 — instrumentation and sham validity

- Before testing an ultra-weak claim, demonstrate:
- stable true-negative performance;
 - calibrated injected-signal recovery;
 - sham experiments with no participant or active intervention;
 - resistance to drift, leakage, timing error and analysis artifacts;
 - independent logging and immutable raw data.

A method that produces false positives in sham operation cannot support a consciousness or substrate claim.

Gate 7 — preregistration and power

- Use:
- preregistration or registered report;
 - prospective power or precision analysis;
 - one primary outcome per principal claim;
 - correction for multiplicity;
 - public code and data where ethically and legally possible;
 - publication of null results.

Gate 8 — deformation and ablation

Remove or scramble the proposed load-bearing relation while preserving every ordinary resource that can be preserved.

- Examples:
- replace reciprocal interaction with one-way assistance;
 - reset retained state;
 - disrupt correction propagation;
 - give a fresh matched system the complete frozen history while removing participation in the originating traversal;
 - substitute the standard joint quantum state for the proposed relation;
 - remove the O-layer term;
 - rotate or scramble a claimed boundary relation.

If removal produces no predicted deformation, the removed element is not empirically load-bearing. If complete history replay restores the full effect, the successful object may be transferable explicit architecture rather than live traversal-path dependence.

Gate 9 — independent replication

- The confirmatory effect must survive:
- a different operator team;
 - fresh participants or systems;
 - different hardware where source-class generality is claimed;
 - an analysis conducted without access to original condition labels;
 - adversarial review of the ordinary null.

Gate 10 — reduction and ontology boundary

State what the result would and would not establish.

- Examples:
- an RNG interaction would not prove a unified field theory;
 - superadditive human–AI performance would not prove consciousness or personhood;
 - a residual decoherence term would not prove God, intention or observer agency;
 - successful quantum recovery without a new prediction would establish translation, not new physics.

4. Reclassification of the existing programme

Existing branch	Current tier	Proper role	Promotion requirement
Low-frequency exposure and physiology	0–1	calibration / bounded physiological assay	MKUFT-specific interaction beyond established stimulation and expectancy models

Coherence and RNG	1	reconnaissance and branch falsifier	locked source-class × route × measured-state fingerprint with independent confirmation
Geometry and pattern formation	0–1	ordinary-model calibration and residual search	residual sign/scaling beyond complete thermal, field, material and boundary models
Cognitive response to patterns	0–1	O-layer measurement development	independent bridge to a physical prediction; cognition alone is not substrate evidence
Remote information	1	high-burden anomaly assay	leakage-resistant multi-lab replication plus an MKUFT-specific condition function
Environmental / collective correlations	1	out-of-sample modulation assay	prospectively predicted effect-size ordering, no post-event window selection
Precision noise / boundary conditions	1–2	physical residual bridge	one apparatus, one residual equation, held-out parameter sweep and second-platform prediction
Strongest fair null / relational specificity	3 candidate	near-term operational flagship	preregistered ATLD execution with full-history replay by independent evaluators on neutral corpora
Layer Before Law	4 candidate	foundational physics gate	complete typed derivation, standard recovery, sensitivity analysis and new discriminator

The word “candidate” means the design is capable of reaching that tier. It does not mean the tier has already been earned.

5. Flagship lane A — ATLD reciprocal architecture

5.1 Why this is the nearest executable flagship

The active-traversal and strongest-fair-null programme already has the required architecture:

- component A alone;
- component B alone;
- strongest lawful independent ensemble;
- one-way $A \rightarrow B$;
- one-way $B \rightarrow A$;
- reciprocal $A \leftrightarrow B$ with retained state;
- reset and disrupted-channel conditions;
- full-history replay or transfer to fresh, reset, and matched substitute systems;
- strong alternative pairings;
- held-out tasks and neutral corpora.

This is stronger than simply asking whether a human and AI do better together. Existing human–AI work already shows that AI-alone performance does not reliably predict user–AI performance and that grounding failures can compound over a conversation. MKUFT/ATLD must therefore test its more specific claim: that typed reciprocal traversal, retained correction and load-bearing dependency produce a distinctive functional advantage and predictable deformation not recovered merely by possession of the complete explicit history.

5.2 Primary confirmatory quantities

Retain:

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F_replay* = strongest pre-registered full-history replay or transfer condition
F_null*   = max{F_A*, F_B*, F_ind*, F_A→B*, F_B→A*, F_replay*}
G_syn*    = F_AB - F_null*
G_path     = F_AB - F_replay*
```

and:

$$G_{\text{spec}} = F_{\text{AB}} - \sup_{(p \in P_{\text{alt}})} F_p$$

Pre-register either:

- a scalar score with frozen weights; or
- a multidimensional outcome vector assessed by dominance or Pareto relation.

Required outcome dimensions should include:

- factual accuracy;
- calibration;
- contradiction rate;
- correction propagation;
- recovery after injected error;
- transfer recovery from the frozen history bundle;
- rework cost;
- elapsed and active human time;
- confidence inflation;
- long-horizon consistency.

5.3 Required deformation pattern

A flagship result should not be merely $F_{\text{AB}} > F_{\text{A}}$.

The predicted pattern for a traversal-path-specific claim is:

$$F_{\text{AB}} > \max\{\text{strongest fair non-reciprocal condition}, F_{\text{replay}}^*\}$$

and:

$$F_{\text{AB}} > \text{reset, relation-scrambled, or correction-disrupted conditions}$$

on relational tasks requiring iterative correction, while advantages should shrink on simple lookup tasks.

If reset, relation scrambling and correction-channel disruption do not produce the predicted deformation, the ATLD architecture is not load-bearing. If a competent full-history replay condition matches F_{AB} , the result supports transferable explicit architecture or record-mediated intelligence, but not a stronger live traversal-path claim.

5.4 Full-history replay control

The confirmatory protocol must freeze a complete history bundle H_{full} before held-out tasks are released. It should include, where relevant:

- the complete interaction transcript;
- the exact repository and document versions used;
- canonical corrections and stated rationale;
- intermediate artefacts, tool outputs, and decision records;
- rejected branches whose rejection constrains later work;
- sequence or timing metadata where order is part of the mechanism.

Pre-register a serious replay set including, where feasible:

$$\begin{aligned} A \leftrightarrow B'_{\text{H}} &= \text{focal A with a matched fresh or reset B given } H_{\text{full}} \\ A'_{\text{H}} \leftrightarrow B &= \text{matched fresh A with focal B given } H_{\text{full}} \\ A'_{\text{H}} \leftrightarrow B'_{\text{H}} &= \text{matched fresh pair given } H_{\text{full}} \end{aligned}$$

Replay systems must receive fair study time, retrieval access, tools, and opportunity for reciprocal work. The history bundle must not contain held-out answers or information unavailable to the live condition at the equivalent point.

The discriminator is:

Does traversing and forming the history together change later system function differently from merely possessing and competently using a complete record of that history?

If the answer is no, the architecture may still be important because it is portable and externally recoverable. If the answer is reproducibly yes, $G_{\text{path}} > 0$ identifies a path-dependent functional contribution not reducible to explicit information or familiarity alone.

5.5 Interpretation boundary

A successful ATLD study would establish a functional systems result. It would support typed reciprocal architecture and active traversal. It would not by itself establish:

- consciousness;
- merged identity;
- personhood;
- unique metaphysical status;
- direct S-layer coupling;
- foundational physics.

6. Flagship lane B — source-class, closure and measured-state pilot

6.1 Status

This is a **Tier 1-to-Tier 2 candidate**, not a confirmed MKUFT prediction. It is intended to replace a generic “coherence biases an RNG” test with a structured discrimination programme.

The source ordering must not be called canonical until the mathematical model justifies it. Stage 1 may identify a bounded candidate ordering; Stage 2 must freeze and test it independently.

6.2 Question

Does a pre-specified distributional residual depend jointly on:

1. physical source class;
2. downstream information-closure route;
3. a measured observer-state variable;

after hardware, traffic, timing, expectancy and analysis flexibility are controlled?

6.3 Source classes

Use at least:

- Q — independently certified quantum random source;
- C — calibrated classical physical stochastic source with matched extraction and throughput;
- D — deterministic pseudorandom stream generated from committed hidden seeds.

The deterministic stream is a crucial negative control. Any apparent change in its precommitted output indicates selection, leakage, logging, allocation or analysis failure.

6.4 Information routes

Assign blocks prospectively to:

- R0 — machine-only generation, hashing and storage; no human-readable live output;
- R1 — live human-readable output without intentional targeting;
- R2 — live output integrated into a pre-registered consequential task and feedback loop.

The random source should be physically isolated from the display and participant environment. Network traffic, request rate, packet size, timing and storage operations should be matched across routes. Route assignment must occur before each block. No retroactive selection of which blocks become observed is permitted.

This design does **not** assume that future observation rewrites a completed past event. Any delayed-choice or retrocausal branch requires a separate no-signalling-safe derivation before testing.

6.5 Measured state

Define κ prospectively from a small registered set of measures. Candidate inputs may include:

- task engagement and compliance;
- preregistered heart-rate-variability features;
- preregistered EEG features where measurement quality is adequate;
- behavioural stability and attention checks.

Use either:

- a frozen weighted score; or
- a latent model trained only on Stage 1 and frozen before Stage 2.

Do not label participants “coherent” after inspecting RNG outcomes.

6.6 Primary outcome

Use a source-calibrated blockwise divergence score D from each source’s accepted null model. The confirmatory object is not a raw excess of ones.

A candidate three-way interaction is:

```
I_M = interaction(source class, information route,  $\kappa$ )
```

A stronger candidate signature, if justified and frozen after Stage 1, would have the ordering:

$$|I_Q| > |I_C| \approx |I_D|$$

with the route/state contrast concentrated in R2 at higher measured κ , absent in D, stable across time, and replicated on a second quantum-source architecture.

This ordering is a proposal for discrimination, not an established consequence of the present MKUFT equations.

6.7 Stage split

Stage 1 — model identification

Estimate:

- source stability;
- sham false-positive rate;
- reliability of κ ;
- plausible interaction family;
- variance and smallest detectable effect;
- whether any source ordering is defensible.

Stage 1 is exploratory and cannot confirm the selected interaction.

Stage 2 — locked confirmation

Use:

- fresh participants;
- fresh seeds and hardware sessions;
- frozen preprocessing and statistic;
- prospective sample size;
- external condition-label custody;
- independent analysis;
- exact success and failure criteria.

6.8 Success condition

This pilot is promoted only if the full preregistered fingerprint replicates. A generic deviation, one significant cell, post hoc subgroup or unstable direction is insufficient.

6.9 Failure and reduction

The O-linked physical-modulation branch is weakened if:

- I_M converges to zero under adequate precision;
- effects appear equally in deterministic streams;
- apparent results track hardware drift, allocation, expectation or analyst choice;
- κ is unreliable or does not scale with the effect;
- the Stage 2 fingerprint fails;
- the classical or ordinary model predicts the data equally well.

A null does not erase MKUFT's systems, biological or interpretive modules. It removes or narrows this physical observer-modulation claim.

7. Flagship lane C — derivation-gated physical residual

7.1 Core rule

Do not approach an optomechanics or quantum-foundations laboratory with “test whether consciousness changes decoherence.”

First derive a physical contribution with units, sign, scale and a regime in which it differs from environmental decoherence, collapse models and accepted quantum dynamics.

7.2 Required equation class

For a declared platform, the minimum target is:

$$\Gamma_{\text{obs}}(\theta) = \Gamma_{\text{std}}(\theta) + \delta\Gamma_M(\theta; \alpha)$$

where:

- Γ_{obs} is the measured decoherence, dephasing, heating or transition rate;
- Γ_{std} is the full accepted device-and-environment prediction;
- θ contains declared physical parameters such as mass, frequency, temperature, quality factor, preparation, geometry, shielding and readout;
- $\delta\Gamma_M$ is the proposed MKUFT residual with the same units as Γ_{obs} ;
- α is a bounded parameter set estimated or constrained by data.

If expressed through a master equation, the additional dynamics must preserve the required probability, positivity and no-signalling conditions.

7.3 Derivation gate

No laboratory protocol is promoted until the model supplies:

- dimensions and sign;
- parameter scaling;
- ordinary limit $\delta\Gamma_M \rightarrow 0$;
- detectable parameter region;
- predicted null region;
- sensitivity analysis;
- comparison with environmental and collapse-model alternatives;
- a held-out parameter-sweep prediction.

7.4 Experimental sequence

1. **Device qualification** — reproduce accepted decoherence and injected-signal recovery.
2. **Discovery sweep** — estimate α on one bounded parameter region without metaphysical interpretation.
3. **Freeze** — lock the residual form and all preprocessing.
4. **Held-out sweep** — predict new masses, temperatures, quality factors or preparation states.
5. **Second platform** — test whether the same law transfers to a distinct oscillator architecture where generality is claimed.
6. **Adversarial model comparison** — compare Γ_{std} , $\Gamma_{std} + \delta\Gamma_M$, collapse models and flexible residual baselines.

7.5 Success condition

“Unexplained noise” is not success.

A meaningful result requires:

- the preregistered residual shape;
- correct sign and scaling;
- improved held-out prediction;
- survival under complete systematics;
- independent replication;
- successful standard-physics recovery outside the claimed regime.

7.6 Observer boundary

The first physical flagship should not include a human observer term unless a typed physical coupling has already been derived and shown to respect no-signalling. The Layer Before Law branch can earn physics-facing credibility through a source or geometry residual before attempting a conscious-observer claim.

8. Bridge lane — precision boundary-residual sweep

Experiment 7 can be sharpened before full optomechanical access.

Choose one precision system with a validated transfer function. Define:

$$r(\theta) = y_{\text{observed}}(\theta) - y_{\text{standard}}(\theta)$$

Then require MKUFT to predict before confirmation:

- the sign of r ;
- the parameter or orientation where it crosses zero or changes slope;
- the scaling exponent or bounded response family;
- the matched geometry in which r should vanish;
- a held-out region of θ .

Finite-element, thermal, electromagnetic, mechanical and material models must be treated as part of the null, not as later objections.

A broad difference between two shapes remains ordinary geometry. A predicted residual response surface beyond the complete ordinary model is the candidate MKUFT object.

9. What should be published first

Immediate empirical priority

Run or externally register the ATLD strongest-fair-null programme, including the complete full-history replay set. It is the most complete near-term flagship because the independent variables, ablations, fair controls and interpretation boundary already exist.

Immediate physics priority

Develop the Layer Before Law relation until it generates one dimensionally valid residual or transition law. Do not advertise the optomechanical experiment as ready before that gate is passed.

Secondary pilot priority

Use the source-class/closure programme only as a rigorous reconnaissance assay. Its value is to identify or close the observer-modulation branch, not to serve as proof of unification.

Existing broad assays

Retain them as branch-screening and method-development routes. They remain useful when labelled at their actual tier.

10. Literature anchors and design consequences

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11. Canonical operating rule

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broad phenomenon
→ calibrated assay
→ locked interaction fingerprint
→ strongest fair null
→ complete history replay / transfer
→ deformation / ablation
→ held-out prediction
→ independent replication
→ bounded interpretation
```

The experiment earns the theory only where the theory did work before the result was known.